

Lab04-Inheritance

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VNU-HCM, HCMUS, FIT, SE

CS202 - Programming Systems

25A01 - 02

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A: YY: 01

H: YY: 05



Students are permitted to use AI assistance for this series of homework.

If you do not use AI: Please include Declaration A in your README.md file: "I did not use AI for this homework."

If you use AI: Please include Declaration B in your README.md file: "I did use AI for this homework." The expected workflow is that you prompt the AI, it generates a plan, you review that plan, the AI executes the action, and you review, edit, and re-prompt as needed. Ultimately, you bear full responsibility for the final result.

Design class diagrams using Mermaid format and implement the required functionality in C++ for the following assignments. For your final submission, please ensure you include the C++ source code, the raw Mermaid syntax, and the rendered image files for each class diagram.

Class diagram: https://en.wikipedia.org/wiki/Class_diagram

Use stereotypes to annotate the meaning of classes and methods.

```
<<interface>>, <<abstract>>, <<concrete>>, <<constructor>>,  
<<destructor>>, <<friend>>, <<static>>, <<virtual>>, <<pure  
virtual>>, <<override>>, <<operator>>.
```

```
Ex: +<<pure virtual>> getSalary(): double
```

Assignment 1 – Company

Company ABC asks you to build an employee management system.

The system should store the following info for each employee: employ id, full name, hire date, address.

There are 2 types of employees in the company: Office employee and worker.

- Salary of an office employee in a month = number of his/her working days * pay rate per day (300.000 VND/day)
- Salary of a worker in a month = number of items he/she produced in the month * rate pay per item (5.000VND/item)

In this week exercise, you are asked to create 3 classes: Employee, OfficeEmployee and Worker. OfficeEmployee and Worker extends/inherits from Employee.

Then you must define and implement the following methods for each Employee classes.

1. Input the information of an employee
2. Print the information of an employee to console.
3. Compute the salary in a month of an employee
4. Define at least 5 constructors for each of 3 classes.

Assignment 2 – Bank Account

Define a class, named BankAccount with the following attributes and methods:

Attributes:

1. Account number
2. Name of owner
3. Social ID of owner
4. Balance (money in the account)

Methods:

1. Input from keyboard.
2. Print out to console.
3. Deposit an amount of money.
4. Withdraw an amount of money. After withdrawing, the balance should be equal or greater than 50.000 VND.
5. Check the current balance.

Then, define another class, named, SavingBankAccount, which extends / inherits from class BankAccount.

The attributes and methods of new classes are below. Attributes:

1. Account number
2. Name of owner
3. Social ID of owner
4. Balance (money in the account)
5. Annual interest rates, for example: 6% in a year (360 days)
6. Period, for example, 1, 2, 3, 6 or 12 months.
7. Number of saving months until now, for examples, 3 months.

Methods:

1. Input from keyboard.
2. Print out to console.
3. Deposit an amount of money. Do not allow the customer to deposit if the number of saving months < period. In fact, the bank teller will ask the customer to create a new saving account to store new amount of money.

4. Withdraw an amount of money. Do not allow the customer to withdraw if the number of saving months < period.
5. Withdraw immediately. Then the saving account is considered as free-period one, and the interest rate is only 2%.
6. Check the current balance. Remember to add the current amount of interest to the balance. Use balance, interest rate, period and number of saving months to compute the interest.
7. Check the interest at this time.

Assignment 3

Draw a class diagram and write a simple console application in C++ to simulate the following screens.

Write a report to explain your solution.

Notes: in this lab, students should focus on products, quantities, unit prices, and the total price. Other features will be the focus of the following weeks.

Assignment 4 – Inheritance in Python

Write a presentation (Markdown format) with at least 20 slides to explain the following OOP concepts in Python:

- Inheritance
- Overriding method in the derived class.
- The order of constructor calls when a derived object is created.

Assignment 5 – Inheritance in Java

Write a presentation (Markdown format) with at least 20 slides to explain the following OOP concepts in Java:

- Inheritance
- Overriding method in the derived class.
- The order of constructor calls when a derived object is created.